Cause-and-Effect (Fishbone) Analysis

Ishikawa Diagrams Using the 6 Ms

CampusSphere – Campus Opportunity Hub

Software Engineering Lab

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Problem 1: Students Miss Opportunities and Deadlines

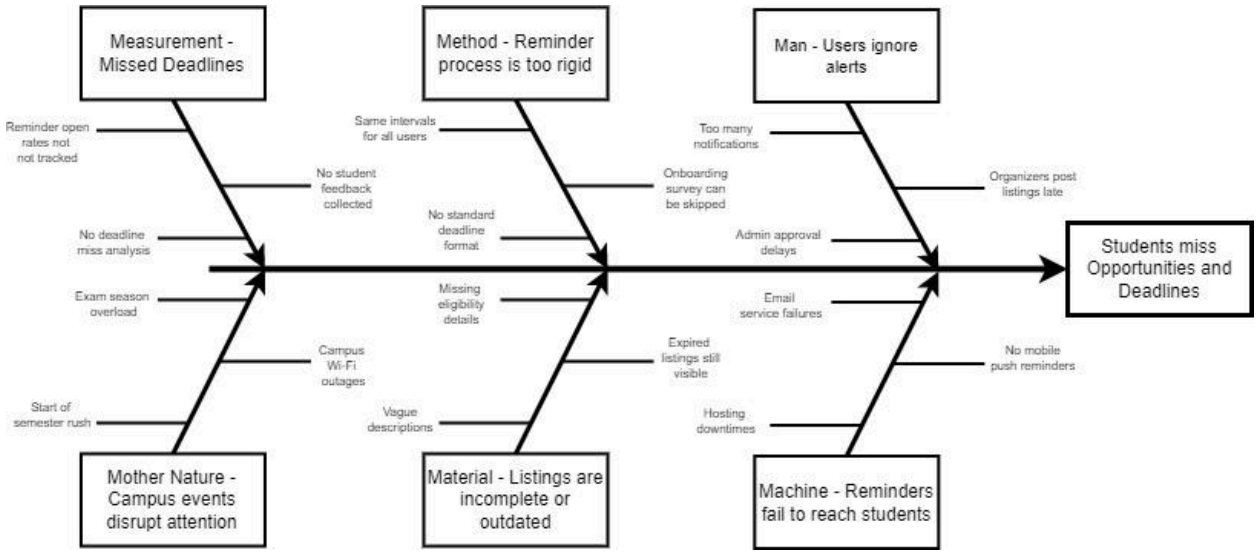


Fig 1 Fishbone diagram for "Students Miss Opportunities and Deadlines"

Explanation

In this diagram, we are looking at the reasons why students still end up missing deadlines and opportunities despite the existence of CampusSphere which ensures that all the listings are brought together under one platform. Man: Students choose not to listen to reminders because they are bombarded by lots of alerts, organizers delay in posting their listings, and approval delays from the admin reduce the period students get to react. Machine: Students do not get reminded because of the email service provider problems, the absence of mobile push reminders, and hosting downtime. Method: The reminder process is not flexible since all users get notified at the same intervals, there is no mandatory onboarding survey, and postings lack any set deadline format. Material: Listings are either incomplete or outdated due to missing eligibility information, expired listings that are still visible, and unclear descriptions. Measurement: Deadlines are never monitored since reminder open rates, student feedback, and deadline misses are never analyzed. Mother Nature: Events in the campus such as exams, Wi-Fi outage, and busy times in the beginning of each semester steal the attention from the platform. Most of these reasons exist in process and data; hence, they need to be addressed, standardized and measured.

Problem 2: Irrelevant Recommendations

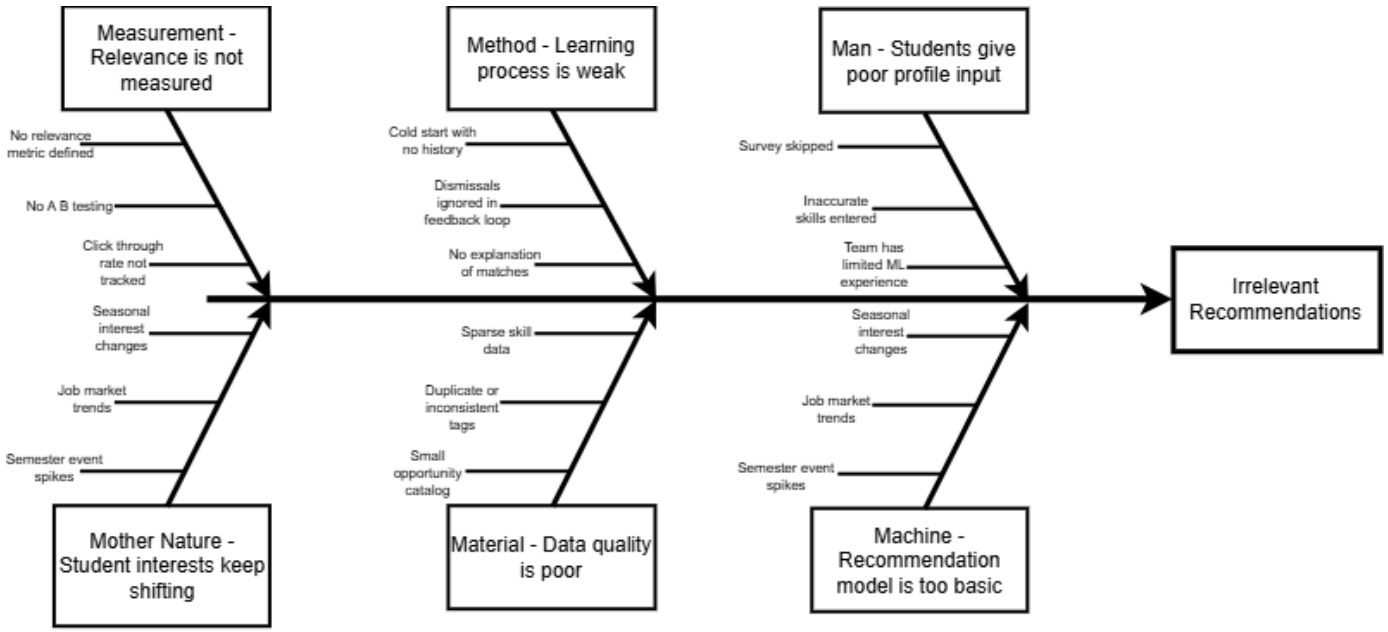


Fig 2 Fishbone diagram for "Irrelevant Recommendations"

Explanation

This diagram shows how it leads to the problem where the recommendation engine recommends opportunities that may not be relevant to the student. Man-wise, students provide poor profile input through not filling out the survey and providing incorrect information for the skills, and the team has little knowledge about machine learning. Machine-wise, the algorithm used in the system is very primitive and uses simple scoring, only refreshes once per day and uses limited computing power from free hosting services. Method-wise, the algorithm has poor learning as the new users do not have any history (cold start problem), dismissals are disregarded during the feedback loop and the matching information is not provided to the user. Material-wise, there is poor data quality where there is a lack of skill information, there are duplications and inconsistencies in the tag information and small number of opportunities in the catalog. Measurement-wise, there is no way to measure the effectiveness of the recommendations as there is no metric for relevance and no A/B testing or click-through rate is tracked. Mother Nature-wise, the interests of the students change according to the seasons, job market trends and semesters.

Problem 3: Slow Response and Downtime at Peak Load

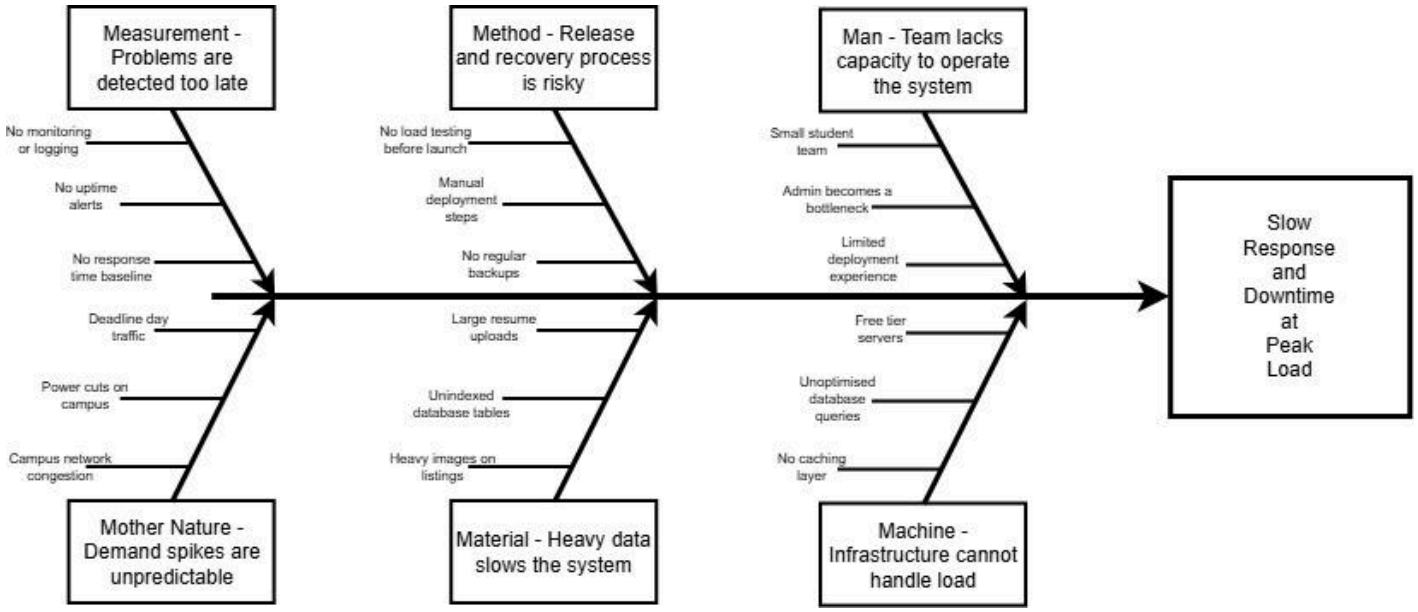


Fig 3 Fishbone diagram for "Slow Response and Downtime at Peak Load"

Explanation

The following diagram highlights the reasons for the slowing down and downtime experienced by CampusSphere under heavy usage, such as during periods leading up to a deadline date. In the context of Man, CampusSphere does not have the capacity to run the system due to it being a small group of students, an admin being a bottleneck and lack of deployment experience. In terms of Machine, CampusSphere lacks the infrastructure capacity as a result of using free tier servers, unoptimized database queries and lack of caching. In regard to Method, the release and recovery process is very risky as there is lack of load testing prior to deployment, lack of automated deployments and lack of backups. With regards to Material, the heavy data slows down the system via large resume uploads, un-indexed database tables and heavy images on the listings. In the case of Measurement, the problems are only detected when it is already too late as there is lack of monitoring/logging, no uptime alerts and no response time baseline.